



Kulkarni Varun Venkatesh
Mechanical Engineering
Indian Institute of Technology Bombay

22B2121
B.Tech.
Gender: Male
DOB: 21/07/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	Maharashtra State Board	Rajarshi Shahu Mahavidyalaya Latur	2022	
Matriculation	Maharashtra State Board	B.W. New English Medium School	2020	

Pursuing a **minor degree** in **Computer Science** from CSE Department, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Awarded an **AP** grade (awarded to <1% students) in **2** core courses along with **AA** grades in **20+** courses (2025)
- Secured **99.74** percentile (**1M+** applicants) in JEE Main & **98.94** (**0.16M+** applicants) in JEE Advanced (2022)
- Recipient of the **NTSE** scholarship granted by NCERT to top **0.2%** (**1M+** applicants) class 10th candidates (2020)
- Achieved a **perfect 100%** score in the 10th Maharashtra SSC Boards examination among **1.5M+** applicants (2020)
- Secured a percentile of **96.25** (**0.1M+** applicants) in the **KVPY SX** exam conducted by **IISc, Bangalore** (2022)

PROFESSIONAL EXPERIENCE

Data Scientist Intern | Microsoft India Pvt. Ltd.

(May '25 - Jul '25)

Received a **Letter of Recommendation** from the Principal Applied Science Manager for exceptional performance

Impact	- Achieved a prediction accuracy of 86% by optimising 3+ specialised classification models - Identified and prioritized top 500 developers to implement targeted retention strategies - Established improvements in the top 3 features leading to a 20% increase in the retention
Approach	- Performed retention analysis on 153k+ developers (705k+ products) on Microsoft Store - Analysed 185M+ data-points with 108 features for 6-month developer activity dataset

Data Analyst Intern | NoQs Digital

(Jun '24 - Jul '24)

Worked on the data accessibility and visualisation; received a **LoR** from the **Founder** for exceptional performance

- Led a team of **5** to deploy a web app using Apps Script, enabling real-time data modification and operational agility
- Improved the accessibility by **200%**, by developing **Power BI** dashboards for analysing **10M+** global sales data

KEY PROJECTS AND COMPETITIONS

Prosperity 3 | IMC Trading

(Apr'25)

A **15-day algorithmic trading** challenge conducted by **IMC Trading** for university students across the world

- Achieved **Global Rank 4** and **India Rank 1** out of the **12,000+** competing teams, advancing through Round 4
- Deployed **pairs trading**, **mean reversion**, and **statistical arbitrage** on synthetic equity and **ETF**-like assets
- Modeled Implied Volatility and locational arbitrage in option-like instruments, generating **863k+** virtual currency

Prediction and EDA on precipitation data | Course Project : AE248

(Feb'24 - May'24)

- Utilised Linear Regression and Neural Network models for **97%** accurate rainfall forecasting from **5-year data**
- Performed **pre-processing** by analysing correlations among **10** variables and eliminating less significant ones
- Validated models via **t-test** and **hypothesis testing** with **99%** confidence interval for robust prediction evaluation

Human Activity Recognition using ML | Course Project : ME228

(Feb'24 - May'24)

- Utilised a **564-featured dataset** to classify human actions into **6** activities based on different physical responses
- Trained **SVC** and **Random Forest Classifier** models on response dataset achieving a test accuracy of **98.5%**
- Evaluated SVC and Random Forest Classifier performances, analysing suitability for the dataset's specific requirements

Social Media Post Generator | Self-Project

(Apr'25 - May'25)

- Automated **content creation** via **multi-agent AI** using **AutoGen** and integrated APIs enabling post generation
- Improved post relevance with LLM-driven **optimisation** of captions, hashtags and visuals for higher **engagement**
- Boosted efficiency through interactive **Streamlit** deployment, delivering ready-to-publish social posts in **real time**

Java Trading Strategy Backtester | Self-Project

(Dec'24 - Jan'25)

- Developed a backtesting engine using **Java** to evaluate different trading strategies in terms of the **Sharpe ratio**
- Validated the engine by testing **SMA** and **EMA** strategies for the top **15 US stocks** over a duration of **10 years**

Options Pricing Models | Winter in Data Science, Analytics Club

(Dec'23 - Jan'24)

- Developed financial models for Options pricing with Neural Networks, surpassing **Black-Scholes** model accuracy
- Optimised MLP prediction models with **20-day historical volatility data** for increased adaptability and accuracy
- Implemented **2** successive **MLP** models, one with the output layer returning the equilibrium price of the option and the other returning the bid and prices and achieved improved accuracy in pricing for **Call** and **Put** premiums

10-K filing Extractor | Self-Project

(Apr'24 - May'24)

- Developed a specialised function to automate the retrieval of **30 years** of **10-K filings** for targeted companies
- Utilized **sec-edgar-downloader** to systematically extract and download 10-K filings from the official **SEC** website

Quantitative Portfolio Management | Summer of Science, Maths & Physics Club (May'24 - Jul'24)

- Learnt the basic mathematics for finance, including Measure Theory, **Stochastic Calculus**, and Convex Optimization
- Applied the **Markowitz Portfolio Optimization** model, evaluated Modern Portfolio Theory and Arbitrage Pricing Theory and conducted a comparative performance analysis for optimal asset allocation and risk-adjusted returns

ENGINEERING EXPERIENCE

Formula Student | IIT Bombay Racing

(Apr '23 - Aug '24)

A **3-tier** cross-functional team of **70+** students involved in the design and fabrication of an electric race car which competes in the Formula Bharat and Formula Student international competition conducted by SAE and IMechE

- **Junior Design Engineer** | Aerodynamics & Bodyworks, Brakes & EBS subsystems (Aug '23 - Aug '24)
 - Orchestrated the **manufacturing, testing, and validation** of the Brakes and Throttle systems used in the current car along with the positioning & implementation of **Brake Over-travel** switch in the brake assembly
 - Improved the design and performance of the **brake disc** using structural and thermal multi-physics simulations
- **Trainee** (Apr '23 - Aug '23)
 - Performed 2D simulations of airflow around an Ahmed Body in **Ansys Fluent** and 3D simulations of airflow around the rear wing of the car in **SimScale** to analyse the flow structure for calculating lift and drag coefficients
 - Gained insights in **Aerodynamics & Bodyworks, Drivetrain, Accumulator and Brake & EBS** subsystems

POSITIONS OF RESPONSIBILITIES

Department Academic Mentor | Student Mentorship Program, IIT Bombay (May '24 - May '25)

Selected in a **43-membered** team out of 150+ applicants through a rigorous process of interviews and peer reviews

- Mentored **8 sophomores** for academics & co-curriculars; guided them through department **curriculum overhaul**
- Responsible for organizing and conducting **technical events** across the department consisting of **200+** students

Marketing Coordinator | The Entrepreneurship Cell, IIT Bombay

(May '23 - Jan '24)

Asia's Largest Entrepreneurship promoting student body recognized by NEN | Patronage from UNESCO

- Incubated, planned, and organized the flagship event of E-Cell, attracting a remarkable footfall of **45,000+** attendees
- Liased with more than **800** professionals on **LinkedIn** and effectively secured the partnership of multiple firms

KEY COURSES UNDERTAKEN

Mechanical Engineering	Engineering Mechanics, Thermodynamics, Solid Mechanics and Strength of Materials, Structural Materials, Fluid Mechanics, Solid Mechanics Lab, Manufacturing Practice Lab, Fluid Mechanics Lab, Applied Data Science and Machine Learning, Mechanical Processing of Materials, Kinematics and Dynamics of Machines
Computer Science	Data Structures and Algorithms, Computer Programming and Utilization, Operating Systems, Computer Networks, Database and Information Systems* (* to be completed by Autumn '25)
Other Courses	Economics, AI and Data Science, Linear Algebra, Differential Equations, Calculus I and II, Introduction to Quantum Physics, Introduction to Classical Physics, Makerspace, Design Thinking for Innovation, Decision Theory and Information Economics

TECHNICAL SKILLS

Languages	C/C++, Python, R, SQL, HTML, DART, Java
Software	Azure, Synapse, MATLAB, Git, Ansys, ABAQUS, ADAMS, SolidWorks, L ^A T _E X
Data Science Libs	Excel, NumPy, Pandas, Matplotlib, SciPy, PyTorch, TensorFlow, PySpark

EXTRACURRICULAR ACTIVITIES

- Participated in **Divisional Level** Inter-school **Boxing** Competition out of **3 districts** (48-50 kg category) ('19)
- **Felicitated** for the amazing performance and dramatic skills in the **Bal Natya Mahotsava Latur '17** ('17)
- Certified by **MGAM** for successfully competing in **Ganit Pradnya Competition** for mathematical skills ('18)
- Completed the Certification course offered by Coursera on **Fundamentals of Quantitative Modeling** ('24)